

USING PGD WITH THE LOADING POSITION AS A PARAMETER TO SIMULATE SURGEON APPLIED LOADS DURING AN EVAR

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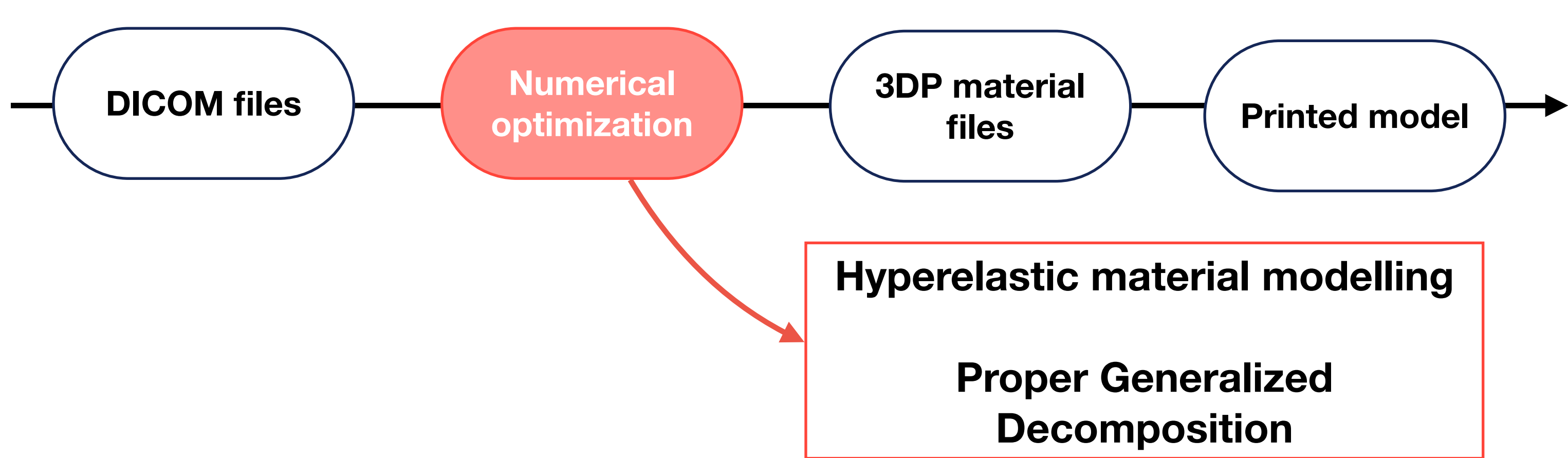
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Model reduction, hyperelasticity, optimization, biomimetic composites

This PhD, a part of the RHU-EndoVx project, is carried out in collaboration with the Hospital Marie Lannelongue (91). Its primary objective is to create realistic synthetic 3D-printed models of abdominal aortic aneurysms (AAA) for surgical training and rehearsals. The work involves the formulation of equations for hyperelastic material models to match the properties of the biomimetic materials used in the construction of the models, as well as the development of an optimization algorithm utilizing Proper Generalized Decomposition (PGD) to accurately simulate their behavior, using the position of the load exerted by the surgeon as a parameter.

Overarching objective: 3D printing of anatomies to help surgeons train and learn the operation

Creating a numerical model for parametric optimization of biomimetic anatomies



PGD using loading position as a parameter

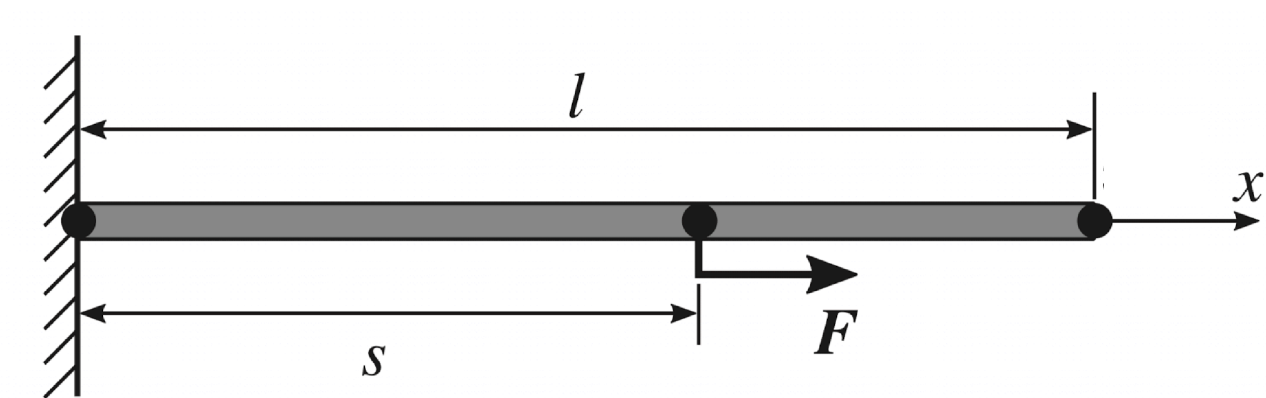


fig. 2 - sketch of the 1D model problem [10]

Parameter definition

Loading position $s \in \Gamma_s$.

Loading type:

- Volume force $\mathbf{f} \Rightarrow \Gamma_s \subset \Omega$
- Surface force $\mathbf{g} \Rightarrow \Gamma_s \subset \partial\Omega$

Aortic aneurysms and surgical interventions

Aneurysm: permanent localized dilatation of an artery having at least a 50% increase in diameter. [6]

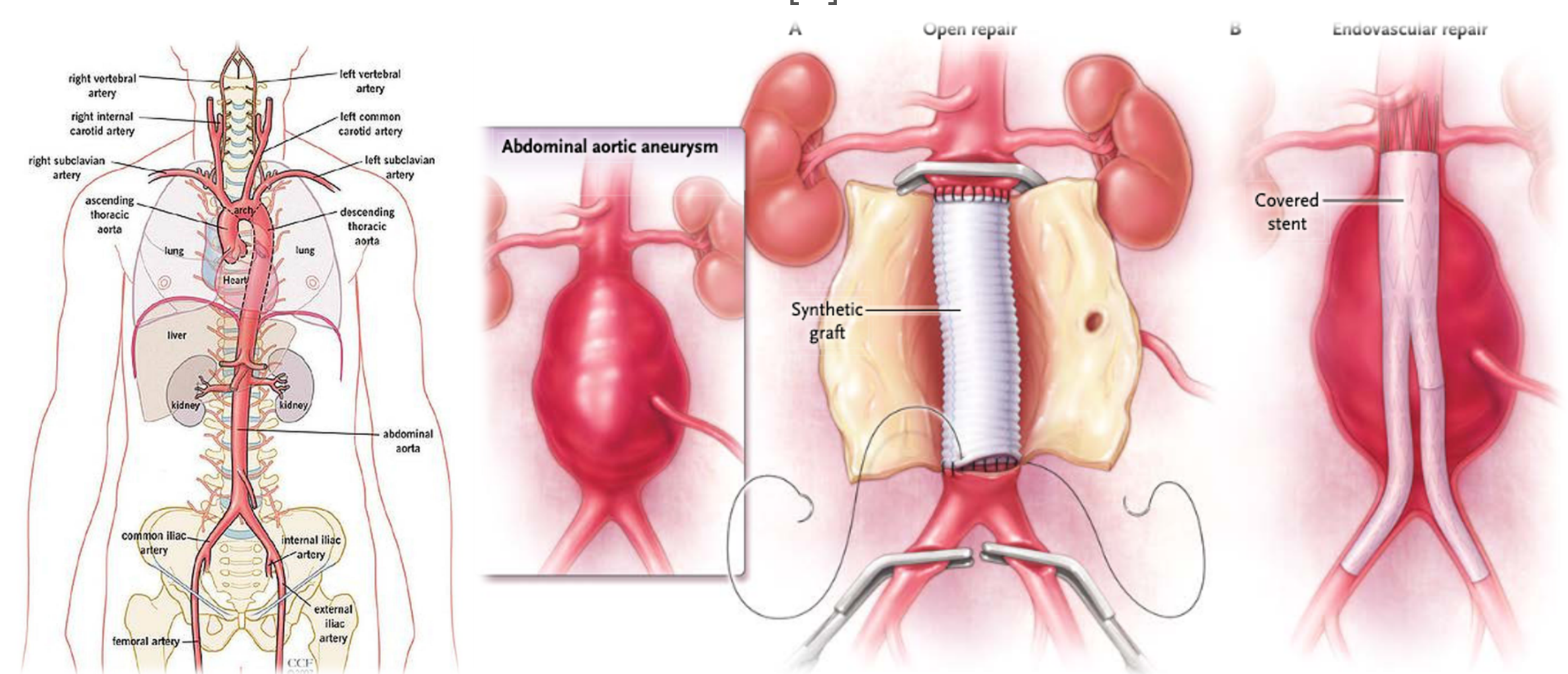


fig. 1 - Anatomy of an aorta in the human body (left), AAA and the two techniques for repair (right)

“Endovascular repair [...] involves the intraluminal introduction of a covered stent through the femoral and iliac arteries; the stent functions as a sleeve that passes through the aneurysm sac, anchoring in the normal aorta above the aneurysm and in the iliac arteries below the aneurysm.” [7]

Linear elasticity

Ω an undeformed domain

$$\nabla \cdot \underline{\underline{\sigma}} + \mathbf{f} = \mathbf{0} \quad \text{on } \Omega$$

$$\mathbf{u} = \bar{\mathbf{u}} \quad \text{on } \partial_u \Omega$$

$$\underline{\underline{\sigma}} \cdot \mathbf{n} = \mathbf{g} \quad \text{on } \partial_\sigma \Omega$$

$$u^n(x, s) = \sum_{i=1}^n \phi^i(x) \lambda^i(s)$$

$\phi^i(x)$: space functions
 $\lambda^i(s)$: parameter functions

Variational formulation

$$a(\mathbf{u}, \mathbf{u}^*) = l(\mathbf{u}^*)$$

Extended PGD variational formulation

$$\mathcal{A}(\mathbf{u}, \mathbf{u}^*) = \mathcal{L}(\mathbf{u}^*)$$

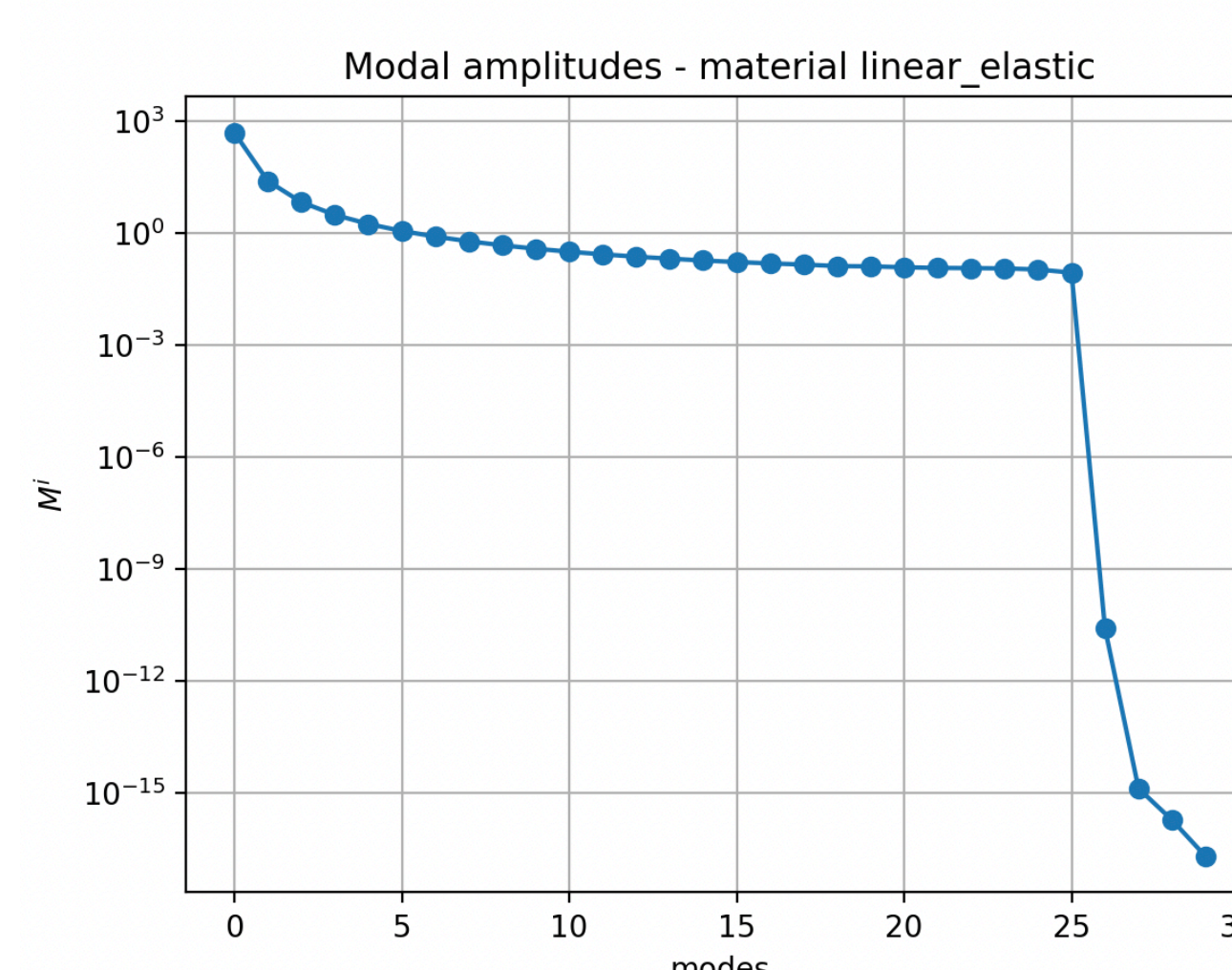
$$\mathcal{A}(\mathbf{u}, \mathbf{u}^*) = \int_{\Gamma_s} \int_{\Omega} \underline{\underline{\varepsilon}}(\mathbf{u}) : \underline{\underline{\varepsilon}}(\mathbf{u}^*) \, d\Omega \, d\Gamma$$

$$\mathcal{L}(\mathbf{u}^*) = \int_{\Gamma_s} \int_{\Omega} \mathbf{f} \cdot \mathbf{u}^* \, d\Omega \, d\Gamma + \int_{\Gamma_s} \int_{\partial_\sigma \Omega} \mathbf{g} \cdot \mathbf{u}^* \, d\partial\Omega \, d\Gamma$$

Standard Galerkin minimisation with a fixed-point algorithm

$$u_{exact}(x, s) = \begin{cases} \frac{Fx}{EA} & \text{for } x < s \\ \frac{Fs}{EA} & \text{for } x \geq s \end{cases}$$

Good fit with analytical solution (error < 1e-3)



$$\forall i \in [1, N_{mod}]$$

$$M^i = \|\phi^i\| \cdot \|\lambda^i\|$$

PGD solution with 30 modes - material linear_elastic

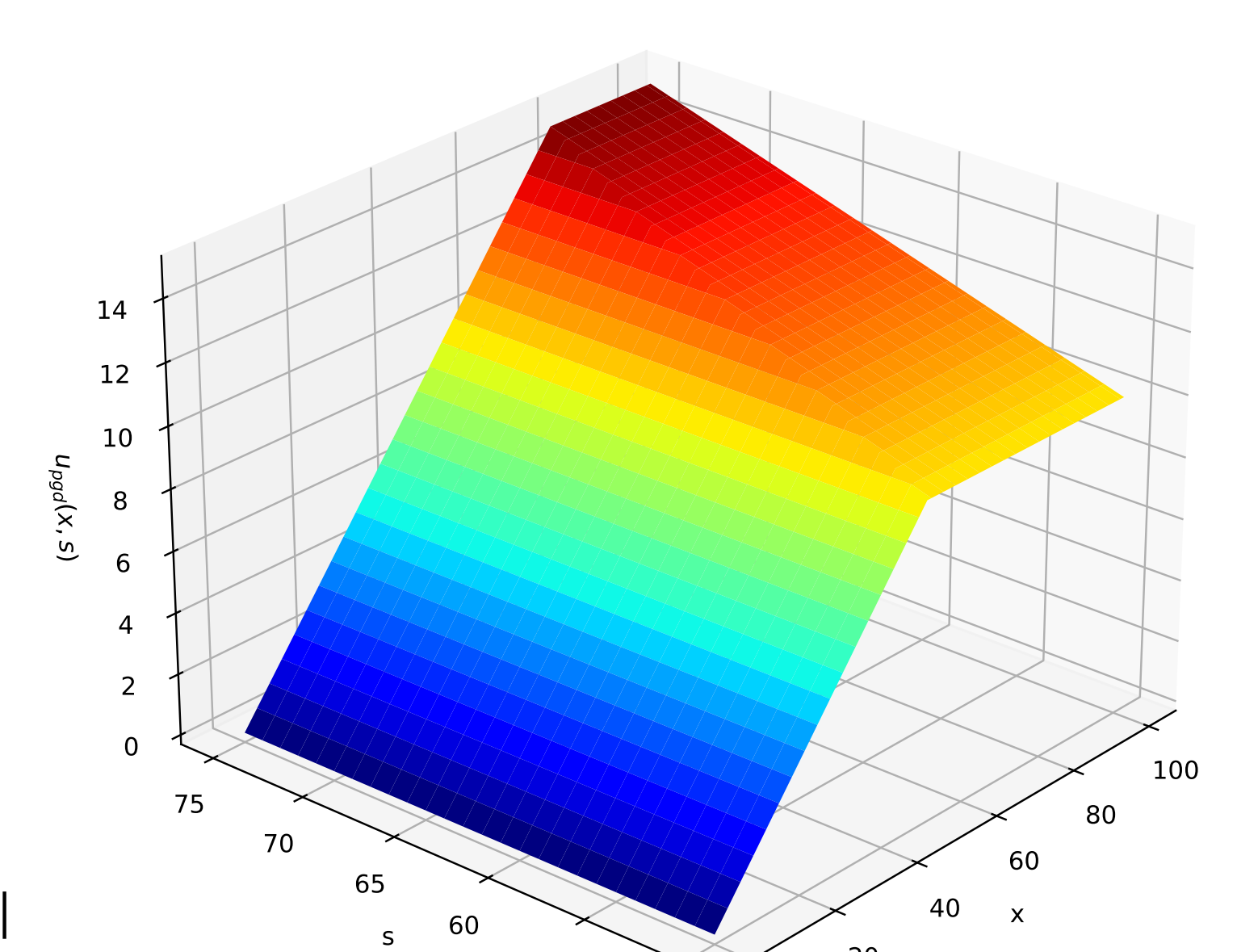


fig. 3 - results for LE with 30 modes: modal amplitudes (left) & solution over the parametric space (above)

Hyperelasticity

Ω an undeformed domain

$$\nabla \cdot \underline{\underline{P}} + \mathbf{f} = \mathbf{0} \quad \text{on } \Omega$$

$$\mathbf{u} = \bar{\mathbf{u}} \quad \text{on } \partial_u \Omega$$

$$\underline{\underline{P}} \cdot \mathbf{n} = \mathbf{g} \quad \text{on } \partial_\sigma \Omega$$

$$\mathcal{A}(\mathbf{u}, \mathbf{u}^*) = \int_{\Gamma_s} \int_{\Omega} \underline{\underline{S}} : \underline{\underline{E}}^* \, d\Omega \, d\Gamma$$

nonlinearity

How to deal with nonlinearity?

Original PGD nonlinear algorithm [3][4] - flexible and lowers computational cost
Applied on NL heat eqn. and NS with convection

Introduction of two linearizations:

- At mode $n = 0$ (and potentially a few select modes): « incremental linearization »
- At all the other modes: « explicit linearization »

Explicit linearization	Incremental linearization
located out of the fixed point loop, in the PGD loop	located in the fixed point loop
takes into account solution from the previous modes only	takes into account solution from the previous modes as well as the previous FP solution
low cost	high cost

